

Part: **3.4 – STRENGTH OF COMPOSITE PARTS**

Chapter: **3495**

Title: **DELAMINATION/UNFOLDING**

Revision: **1.0**

Date: **27/10/2021**

## 1 INTRODUCTION

### 1.1 METHOD-SUMMARY

This chapter presents the methodology proposed to perform the analysis of composite material laminates, subjected to significant interlaminar stresses.

#### Laminate beams

This is the case of plane or curved laminates, subjected to significant out-of-plane applied loads. The analysis proposed is based in beams theory (straight and curved), and covers the calculation of the interlaminar stress and the static margin for the delamination failure.

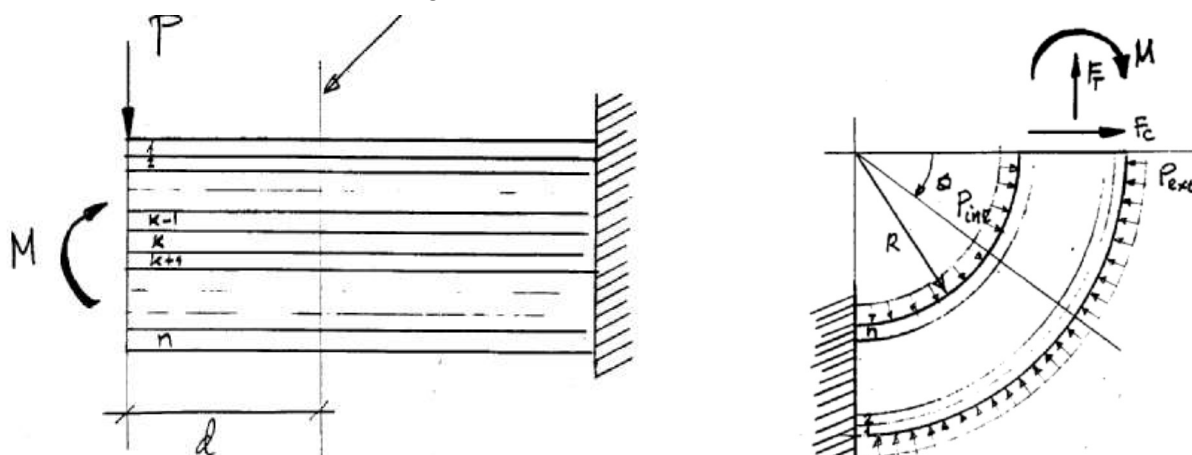


Figure 1. Laminated beam models (straight and curved) in delamination/unfolding analysis

As noted above, the method is based in beams theory, the problem constrained within one plane (through-thickness) and unable to capture tri-dimensional effects. Nevertheless, this is not a

significant limitation in practice for conventional composite parts designs, where interlaminar effects due to direct out-of-plane loads applications are typical contained in one plane (that normal to the web-to-flange folding axis, near a bolted attachment).

#### Free-edge effects

This problem accounts for the interlaminar shear stresses induced at free edges in composite laminates when subjected to in-plane loads.

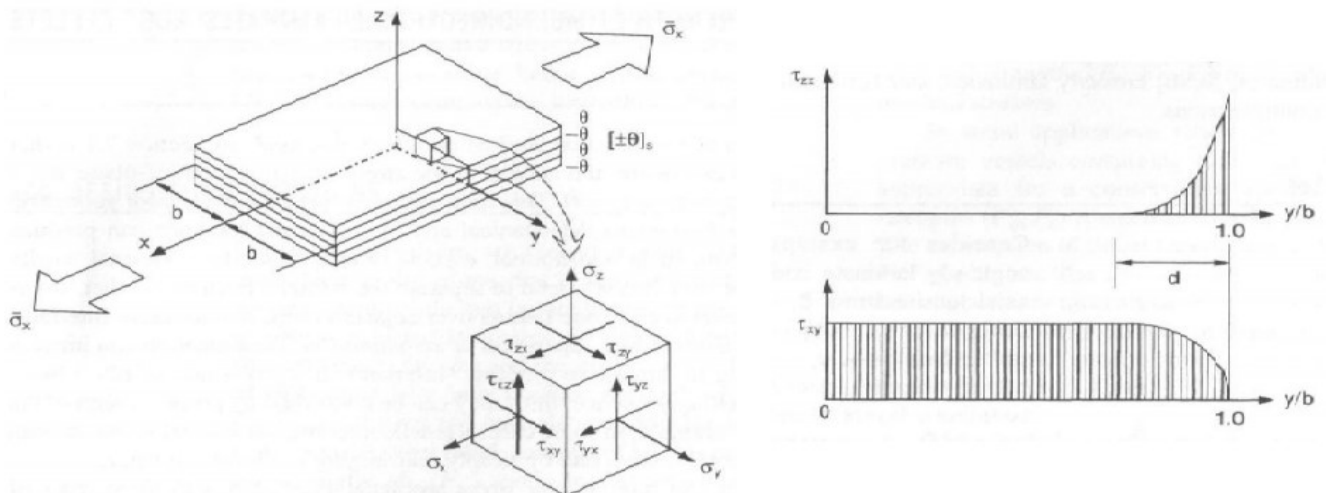


Figure 2. Interlaminar stresses at free edges

Dedicated stress/strength calculations for this detail shall not be frequent or necessary, but restricted to special circumstances (as well as for the case of tapered laminates, see Ref. 4 and Ref. 5).

## 2 METHOD DESCRIPTION

Only a high level overview of the method shall be presented here, being the mathematical details too dense to be presented here and sufficiently described in the documents included in the list of REFERENCES.

### 2.1 STRESS ANALYSIS

#### 2.1.1 Laminated beams

##### Laminated Straight Beams

The problem statement is graphically described in the first of the figures in Figure 1. Analysis method and solution is presented in Chapter III, section 18 of Lekhnitskii's book (Ref. 2). Further details of Airbus DS method to get in-plane and interlaminar shear stresses can be found in Ref. 1.

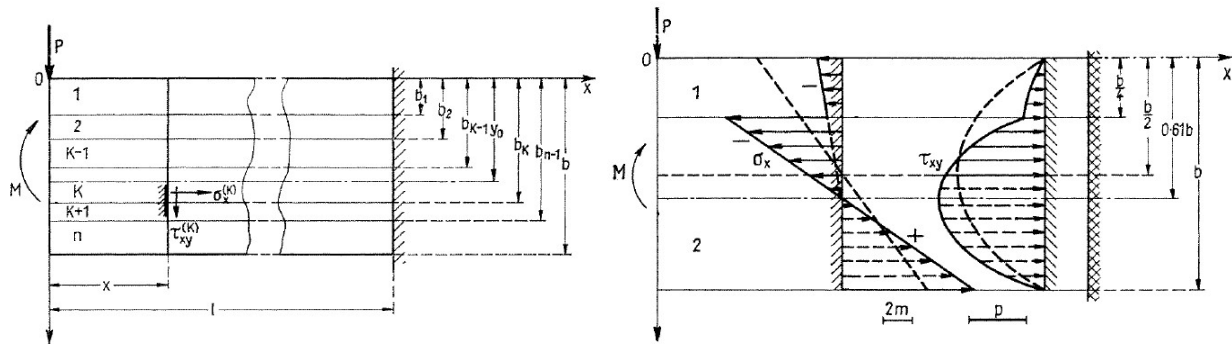
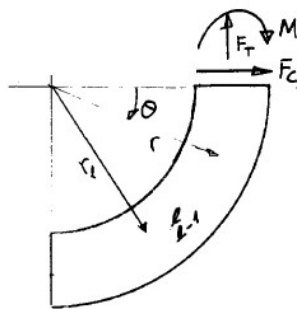


Figure 3. Figures 28 (analysis model) and 30 (calculation example) of Lekhnitskii's book

### Composite Folded Angles

For this case, the problem statement is graphically described in the second figure in Figure 1. This case is of particular interest for the strength analysis of folded composite angles fastened to other parts when attachment bolts/rivets must sustain tension loads (ribs to covers in lift boxes, or frames to skins in fuselages, for example).



$$\frac{\partial \sigma_r}{\partial r} + \frac{1}{r} \frac{\partial \tau_{r\theta}}{\partial \theta} + \frac{\sigma_r - \sigma_\theta}{r} = 0$$

$$\frac{\partial \tau_{r\theta}}{\partial r} + \frac{1}{r} \frac{\partial \sigma_\theta}{\partial \theta} + 2 \frac{\tau_{r\theta}}{r} = 0$$

Figure 4. Analysis model and equilibrium equations in polar coordinates

The analysis method states the equilibrium equations of the curved beam, introduce some prescribed stress functions compliant with those equations and imposes compatibility of deformations at layer interfaces for adjusting the parameters of those shape functions and obtain the solution.

Note that, in curved beams, the radial stresses, in addition to interlaminar shear, play a relevant role in the delamination/unfolding problem.

With limitations (homogenous laminate, moderate  $r_i/r_o$  ratio ...), and under certain circumstances, the simple formulas for the maximum interlaminar shear stress in plates, and for the maximum radial stress in curved beams, as provided by literature, which are used, for example, in deriving the material allowable *ILSS* and unfolding strengths in composites (see Ref. 2, Ref. 6, Ref. 7), could serve as practical approximations:

- ✓ Maximum shear stress  $\tau = 1.5 \cdot Q_z / t$
- ✓ Maximum radial stress  $\sigma_r = 3 \cdot M / [2 \cdot t \cdot (r_i \cdot r_o)^{0.5}]$

### 2.1.2 Interlaminar stress at free edges

For a flat straight laminate under axial load parallel to the free-edge (case presented in Figure 2), the interlaminar stresses can be computed using the method proposed by the ESDU 95028 (Ref. 4).

## 2.2 FAILURE CRITERIA

### 2.2.1 Delamination (out-of-plane) problem

A quadratic failure law is proposed to combine failure indices for the radial-tension and shear stresses in order to get the combined static margin for the delamination failure:

- If radial stress  $\sigma_r$  is positive (tension):  $RF = 1 / [ (\tau / ILSS)^2 + (\sigma_r / F_{3t})^2 ]^{0.5}$
- If radial stress  $\sigma_r$  is negative (compression)  $RF = ILSS / \tau$

### 2.2.2 In-plane failure

Refer to chapter 3403 "PLAIN STRENGTH OF COMPOSITE LAMINATES".

## 3 APPENDICES

### ABBREVIATIONS AND SYMBOLS

| <i>Symbol</i> | <i>Units</i> | <i>Definition</i>                                                                                                                                                 |
|---------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $F_{33t}$     | MPa          | Unfolding ultimate stress of the material                                                                                                                         |
| $ILSS$        | MPa          | Interlaminar shear ultimate stress of the material (also referred as $F_{13}$ , $F_{23}$ for differentiating longitudinal and transverse shear planes in the ply) |
| $N_x, F_T$    | N/mm         | In-plane tensile load in the laminate                                                                                                                             |
| $Q_z, F_C$    | N/mm         | Through-thickness shear load in the laminate                                                                                                                      |
| $M_x, M$      | N            | Bending moment in the curved laminated beam, positive if causing tension tangential stress at the inner face                                                      |
| $i$           | -            | As subscript, relative to the most inner face of the curved beam                                                                                                  |
| $o$           | -            | As subscript, relative to the most outer face of the curved beam                                                                                                  |
| $t$           | mm           | Thickness of the laminate                                                                                                                                         |
| $r$           | mm           | Radial coordinate/distance                                                                                                                                        |
| $\vartheta$   | -            | Circumferential coordinate/angle                                                                                                                                  |
| $\sigma$      | MPa          | Axial stress                                                                                                                                                      |
| $\tau$        | MPa          | Interlaminar shear stress                                                                                                                                         |
| TBC           |              | To Be Confirmed                                                                                                                                                   |

### REFERENCES

|        | <i>Document Reference</i> | <i>Issue</i> | <i>Title</i>                                                                                                                   |
|--------|---------------------------|--------------|--------------------------------------------------------------------------------------------------------------------------------|
| Ref. 1 | NT-T-ADCE-018/92          | Ed. A        | MANUAL TEORICO DEL PROGRAMA CORIIN, Versión 1.0                                                                                |
| Ref. 2 | [S. G. Lekhnitskii]       | 1984         | ANISOTROPIC PLATES                                                                                                             |
| Ref. 3 | [S. P. Timoshenko]        | 1959         | THEORY OF ELASTICITY                                                                                                           |
| Ref. 4 | ESDU 95028                | 1995         | Delamination and free edge stresses in composite laminates subjected to uniform prescribed axial strain and temperature change |
| Ref. 5 | ESDU 91003                | 1991         | Delamination of tapered composites                                                                                             |
| Ref. 6 | ASTM D2344/D2344M         | -16          | Standard Test Method for Short-Beam Strength of Polymer Matrix Composite Materials and Their Laminates                         |
| Ref. 7 | ASTM D6415/D6415M         | -06a         | Standard Test Method for Measuring the Curved Beam Strength of a Fiber-Reinforced Polymer-Matrix Composite                     |

#### SOFTWARE

| <i>Program</i> | <i>Version</i> | <i>Description</i> |
|----------------|----------------|--------------------|
|                |                |                    |

Table 1. *Software programs referenced in this chapter*

#### RECORD OF REVISIONS

| <i>Revision Date</i> | <i>Authors</i> | <i>Change Reason</i> | <i>Pages/Sub-chapters affected</i> |
|----------------------|----------------|----------------------|------------------------------------|
| 1.0<br>29/09/2021    | G. Baños       | First issue          | All pages                          |
|                      |                |                      |                                    |

Table 2. *Record of Revisions: authors, change reasons and sections/pages affected*

#### APPROVAL SIGNATURES TABLE

| <i>Dpt. \ Site</i>                | <i>Getafe</i>          | <i>Manching</i>        |
|-----------------------------------|------------------------|------------------------|
| <i>Stress Methods (TEPAP-TL4)</i> | G. Baños<br>29/09/2021 | F. Glock<br>22/10/2021 |

Table 3. *Approval signatures table for last revision of this chapter*